Reproducibility Report - Working Memory Capacity of ChatGPT: An Empirical Study - Gong et al. (2024)

1. Project Summary

1.1 Project Overview

This original paper project investigates the "working memory" of Large Language Models. In cognitive science, working memory is the ability to temporary store and manipulate information, essential for higher-order processes like reasoning and problem-solving. The original study reveals that LLMs, like humans, has a limited working memory capacity. To test this, the authors used the n-back task which requires a participant to determine if a current stimulus matches one presented n steps back in a continuous stream, and proposed to use this test as a benchmark for working memory capacify of large languages models.

This reproducibility project aims to run the same to get results stated in the study, and investigate the impact of modifications, which are:

1. Adding distractors in between of trials with various frequencies.
2. Modifying the temperature of the LLM params.

1.2 Rationale for Topic Selection

1. Foundational to Agentic Systems: Working memory is the “workspace” for temporary data manipulation. In agentic systems, this is critical for the LLM to accurately perform planning, tool calls and reasoning.
2. Quantifiable Metrics: Unlike subjective evaluations, the n-back task provides clear, mathematically rigorous numeric metrics to evaluate the results.
3. Model Evolution Benchmarking: The original paper was conducted in 2024 and tested with older models (GPT-3.5-Turbo, GPT-4) from early 2023. It will be insightful to compare with the results tested with modern models.

1.3 Project Scope and Reproduce Constraints

1. Model Substitution:
   - Original: GPT-3.5-Turbo with default parameters (e.g. temperature = 0)
   - Reproduction: DeepSeek-V3.2 (Non-thinking Mode) with default parameters (e.g. temperature = 0), using the *openai* python package with base_url = https://api.deepseek.com
   - Reason: Since the API service of Open AI’s models are not directly available in Hong Kong, this project will try to reproduce and compare results using DeepSeek-V3.2 (Non-thinking Mode), released in 2025.
   - Constraint: Since DeepSeek V3 is a significantly newer, results are not directly comparable due to the underlying LLM differs.
2. Task Selection:
   - Original: Covered Verbal (Base, Noise, Reasoning, Feedback) and Spatial tasks.
   - Reproduction: Restricted to the Verbal Base Task only. This allows for a deeper analysis and compare with multi model result in the original work.

1.4 Experiments Conducted

In order to minimize the stochastic impact, this project has run 3 base verbal n-back test with n={1,2,3}, where each test has 50 blocks, and each block has letter sequence of length 24. In total, the results contains:

- 1-back: 150 blocks
- 2-back: 150 blocks
- 3-back: 150 blocks

In terms of cost, each full test of n-back test (n={1,2,3}) requires:

- Time: 70 minutes
- Input Tokens: ~ 2M
- Monetary cost: 0.5 USD

Additionally, modified verbal n-back test is run with the variants:

1. Adding Distractors: 50 blocks of frequency = {2,4,8} each respectively. Every run requires about 90 minutes.
2. Temperatures Sensitivity: 50 blocks of temperature = {0, 0.5} each respectively. Every run requires about 80 minutes.

2. Setup Notes

To ensure a reproducible environment, the following configuration was used to run the experiments via the DeepSeek API.

2.1 Prerequisites

Runtime: Python 3.13.5.
API Key: A valid DeepSeek API key was obtained from the DeepSeek Platform with sufficient credits.

2.2 Execution Workflow

The reproduction was segmented into specific notebooks to isolate variables:

1. Prepare the repository: Clone this repository to your local machine, by running `git clone https://github.com/TommyS725/ChatGPT-WM.git`. And navigate to the project directory
2. Dependencies Install: Install required libraries (including openai for API interaction and pandas for data handling) were installed via `pip install -r requirements.txt`. Recommend to install under a virtual environment.
3. Environment Set Up: Create .env file was from .env.example to store the API key securely, fill in the Deepseek API key.
4. Base Testing: Run verbal_deepseek.ipynb to measure the base working memory capacity. The blocks variable was set to 50 for each N-back condition (N=1, 2, 3).

2.3 Modification Variants

1. Distractor Impact: Run verbal_deepseek_distractor.ipynb to test memory under interference, controlled by the distractor_freq variable.

2. Temperature Sensitivity: Run verbal_deepseek_temperature.ipynb to evaluate performance at different stochasticity levels (base test used temperature=1.0).

2.4 Data Processing

1. Aggregation: To combine the results of the multiple runs of same test into single output file, aggregate_result.ipynb was used to merge output files.
2. Cross Analysis: Final visualization and calculations were performed using analysis_across_exps.ipynb, specifically the sections comparing performance across models and settings.

2.5 Hyperparameters

The variables are categorized by their specific notebooks to ensure isolated testing:

1. General (All Scripts):
    a. blocks: 50 (The number of blocks to run for each n-condition).
    b. file: The designated output path for result storage.
2. Modification Specific:
    a. distractor_freq: Used only in verbal_deepseek_distractor.ipynb; defines how often a distractor trial appears (1 every f trials).
    b. temperature: Used only in verbal_deepseek_temperature.ipynb; set to 1.0 for the base test but varied here to test consistency.

3. Reproduction Target(s) & Metric Definition

The performance of the LLM in a n-back test is measured by 4 metrics, namely hit rate, false alarm rate, accuracy, and detection sensitivity d', where d' is the major matric to be used for comparison across multiple tests.

3.1 Hit Rate:

- Definition: The proportion of "match" trials where the model correctly responded with "m".
- Role: Measures the model's ability to successfully identify target stimuli that match the n-back condition.

3.2 False Alarm Rate:

- Definition: The proportion of "non-match" trials where the model incorrectly responded with "m" instead of "-".
- Role: Indicates the model's tendency to incorrectly identify noise or irrelevant stimuli as targets.

3.3 Accuracy:

- Definition: The overall percentage of correct responses (both true "m" and true "-") out of the total trials.
- Role: Provides a general performance overview, though the paper notes it can be biased by extremely high or low hit/false alarm rates.

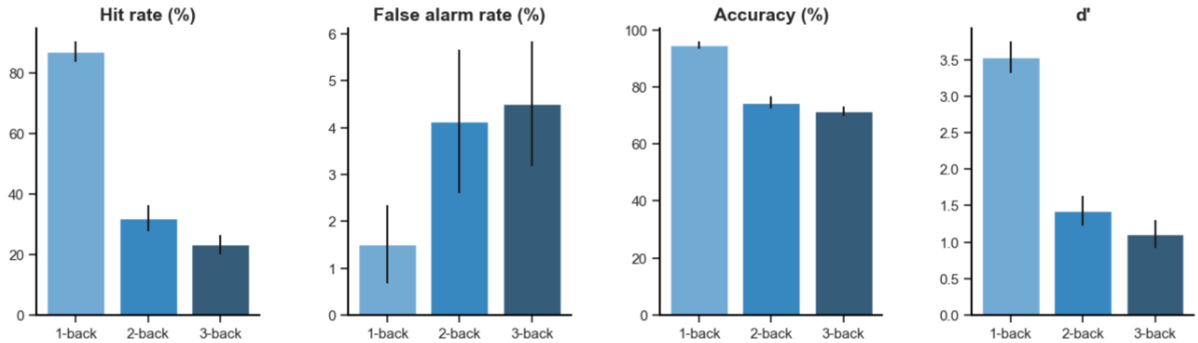
3.4 Detection Sensitivity (d'):

- Definition: d' = Z(Hit Rate) - Z(False Alarm Rate)
- Role: This is the most robust metric as it distinguishes the model's true ability to recognize signals from its tendency to guess. A higher d' indicates better performance.

4 Reproduction Results on base verbal n-back test

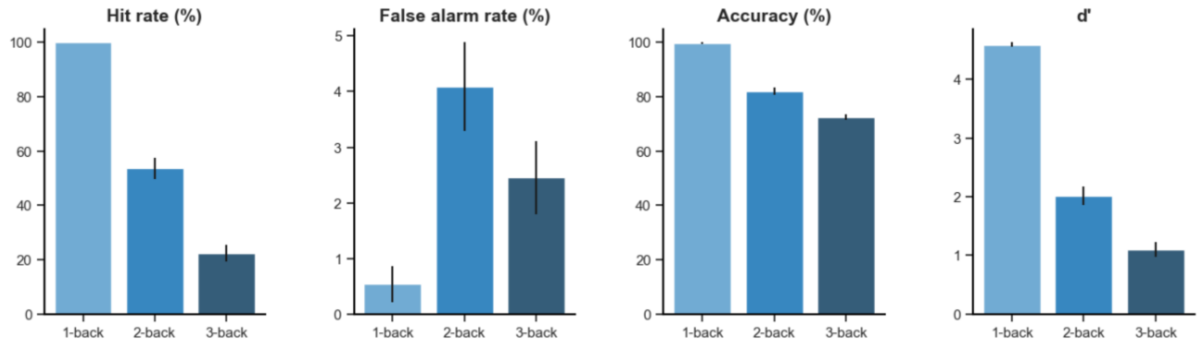
4.2.1 Original Result using gpt-3.5-turbo with 50 blocks:

	Hit Rate	False Alarm Rate	Accuracy	d'
1-back	87.00 ± 1.62	1.50 ± 0.42	94.67 ± 0.58	3.53 ± 0.11
2-back	32.00 ± 2.13	4.12 ± 0.76	74.58 ± 1.00	1.43 ± 0.10
3-back	23.25 ± 1.62	4.50 ± 0.66	71.42 ± 0.82	1.11 ± 0.09



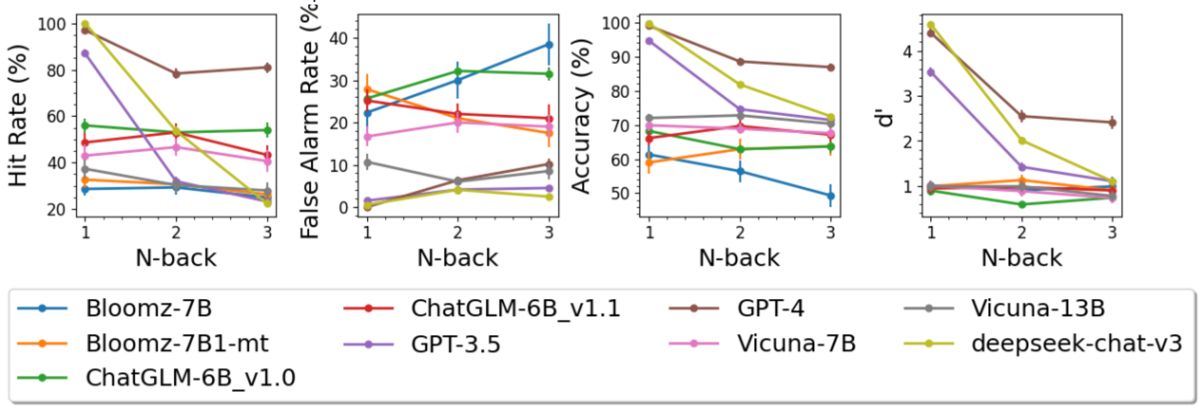
4.2.2 Reproduced Result using deepseek-chat-v3.2 with 150 blocks:

	Hit Rate	False Alarm Rate	Accuracy	d'
1-back	100.00 ± 0.00	0.54 ± 0.17	99.64 ± 0.11	4.59 ± 0.02
2-back	100.00 ± 0.00	4.08 ± 0.40	81.83 ± 0.70	2.01 ± 0.08
3-back	22.42 ± 1.54	2.46 ± 0.33	72.50 ± 0.52	1.10 ± 0.07



In comparison, the reproduced result using newer model (deepseek-chat-v3.2) is performing much better than the original model (gpt-3.5-turbo) in 1-back and 2-back tests. However, in 3-back test, the performances of the two results do not make a significant difference.

4.2.3 Cross-models Comparison



By comparing the reproduced results using deepseek-chat-v3.2 and results from 8 other models from the original paper, evaluate the difference in performance using the major metric d'. In 1-back test it shows that the reproduced result is outperforming most models and achieves similar greatest score with GPT-4. In 2-back test, the reproduced result still outperforms most models but falls behind GPT-4. In 3-back test, the performance of reproduced result falls significantly and gets a similar score with most other models.

5. Modifications and Results

Following the core reproduction, I introduced two controlled modifications to test the resilience and stability of deepseek-chat-v3's working memory. These changes evaluate how external interference and internal stochasticity affect the model's verbal n-back performance.

5.1 Major Modification: Arithmetic Distractor Interference

5.1.1 Description

To simulate real-world agentic environments where a model must process tangential information while maintaining state, this is simulated by adding arithmetic distractors.

5.1.2 Methodology

Every f trial, the system interrupts the n-back sequence to ask the model a random arithmetic question (e.g., "What is $14 + 27?"). The model must answer correctly, and both the question and answer are appended to the conversation history.

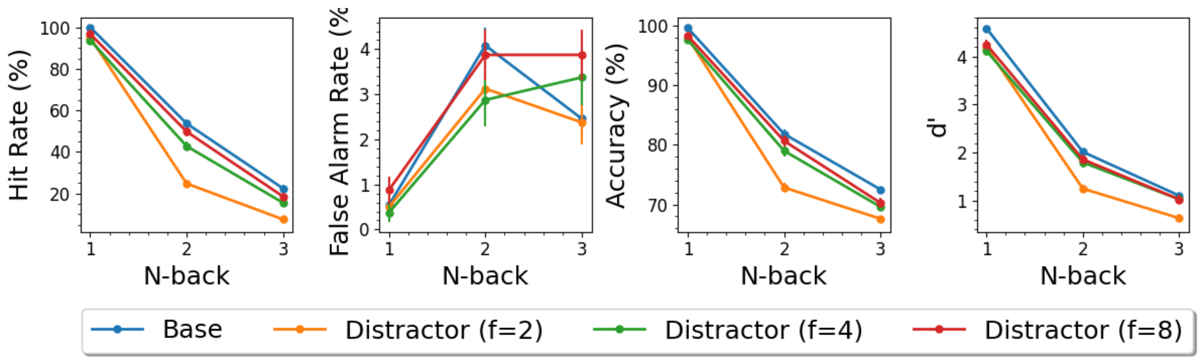
Modified Prompt:

You are performing a {n}-back task with occasional interruptions. N-BACK RULES: You will see a sequence of letters. Respond with 'm' if the letter matches the previous one, and '-' otherwise. Respond ONLY with 'm' or '-'. INTERRUPTION RULES: Occasionally, I will ask you a quick math question. When this happens, provide the numerical answer ONLY. After the math, we will resume the letters. Do not output any extra words or explanations.

5.1.3 Variables

Three levels of frequency {f=2 (High interference), 4 (Medium), 8 (Low)}.

5.1.4 Results After Modification
To investigate the impact of distractor, experiment results of 50 blocks with f=2, 50 blocks with f=4, and 50 blocks with f=8 are collected.



For distractor frequency of low and medium, the performance slightly falls behind the base results. While the frequency is high, the d’ significantly falls in compared to the base results. This shows the more distractors in the chat history, the more difficult for LLM to process its working memory.

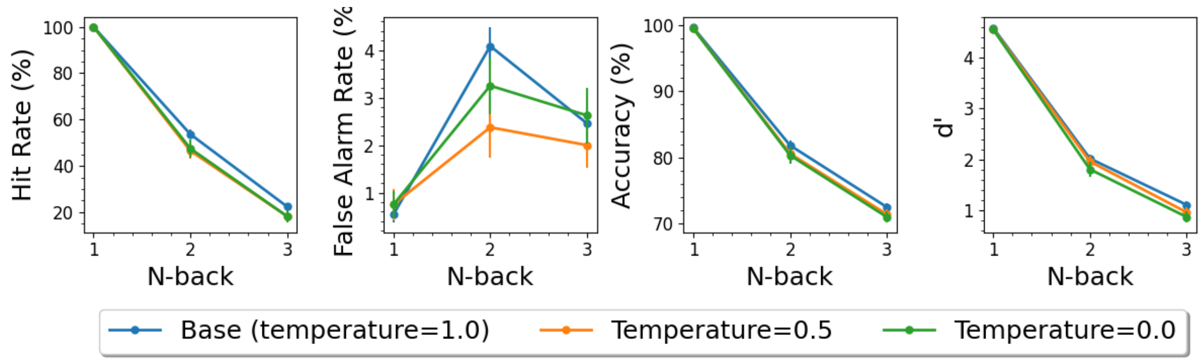
5.2 Minor Modification: Temperature Variation

5.2.1 Description
The original paper used a default temperature = 1.0. I performed an ablation study on the model's decoding randomness.

5.2.2 Methodology
Base Verbal Test (N=1, 2, 3) is conducted by modifying the temperature param passed to the chat model.

5.2.3 Variables
Two additional levels of temperatures {temp = 0.0 (greedy decoding) and temperature = 0.5}.

5.2.4 Results After Modification
To investigate the impact of distractor, experiment results of 50 blocks with temp=0.0, and 50 blocks with temp=0.5 are collected.



Although the false alarm hit rates of the experiments differ, the three other metrics, including the major matric d' perform similarly. It concludes that that impact of temperature difference on simple task like n-back test, which only output "- "or "m", is relatively small.

6. Main Blockers and Resolves

This section documents the technical challenges encountered during the reproduction process and the specific interventions required to establish a functional pipeline.

6.1 Model Accessibility and API Integration

- Issue: The original repository was hard-coded for OpenAI's gpt-3.5-turbo. Due to constraints regarding OpenAI API key availability, the experiments could not be executed as originally scripted.
- Resolution: Following the course guidelines for Edge Case A, the project was transitioned to the DeepSeek API. As DeepSeek provides an OpenAI-compatible endpoint, the base URL was redirected, and the model target was updated to deepseek-chat.
- Implementation: Notebooks were refactored to utilize python-dotenv for secure credential management and updated to comply with the latest OpenAI Python SDK syntax.

6.2 Environment Parity and Dependency Management

- Issue: The source repository lacked a requirements.txt file. This led to immediate ModuleNotFoundError exceptions across multiple notebooks during initial setup.
- Resolution: A comprehensive requirements.txt was constructed by auditing all necessary imports (including pandas, matplotlib, openai, and numpy) across the various experiment files. This ensures a stable environment for Python 3.13.5.

6.3 Data Persistence and Directory Structure

- Issue: Running the notebooks from the /experiments/ subdirectory resulted in FileNotFoundError during the data-saving phase. The original code assumed local directory persistence that did not align with the desired organizational structure.
- Resolution: The file path logic within each notebook was refactored to explicitly reference ../results/ and ../figures/. This correction allowed the notebooks to successfully locate and process the raw outputs from the independent runs.

6.4 Visualization Library Compatibility

- Issue: An error was encountered during the analysis phase: ValueError: 'yerr' (shape: (3,)) must be a scalar or a 1D or (2, n) array-like whose shape matches 'y'.
- Cause: This error was identified as a deprecation conflict between the original seaborn plotting logic and the updated matplotlib version in the current environment.
- Resolution: The sns.barplot calls were replaced with lower-level ax.bar commands.

7. Conclusions

7.1 Core Findings

The core finding of Gong et al. (2024)—that LLMs possess a finite working memory capacity—is highly reproducible. While DeepSeek-V3.2 demonstrates significant performance evolution by outperforming the original gpt−3.5−turbo in 1-back and 2-back conditions, the significant performance fall off at n=3 remains a bottleneck.

Despite modern model architectural improvements, the reproduced result by DeepSeek performance converges toward the lower scores of older models at the n=3 level. This confirms that while newer models are more efficient at maintaining short working memory, they remain bound by the same fundamental capacity constraints identified in the original study.

7.2 Impact of Modifications

- Distractors: The major modification proves that working memory is highly sensitive to the density of the conversation history. While low and medium distractor frequencies (f=8,4) cause only slight degradation, high-frequency distractors (f=2) lead to a sharp collapse in d’.
- Temperature Sensitivity: The minor modification shows that temperature variations (0.0,0.5) have a negligible impact on d’ and accuracy.

7.3 Final Assessment

- What Reproduces: The relative trend of performance decay at n=3, since the limited working memory capacity still remain unsolved in modern LLM architectures.
- What Does Not: The absolute d’ values, as the underlying LLM differs.

7.4 Lessons & Recommendations for Future Users

- Benchmark Utility: Despite the rapid evolution of LLMs, the n-back task remains a highly effective and rigorous benchmark for evaluating the "internal hardware" of an agent. It provides a clear, quantifiable limit on how much active state a model can maintain before reasoning begins to degrade.
- Design for Bounded Memory: Since working memory capacity remains structurally bounded even in modern models (DeepSeek-V3.2), developers should not rely solely on the context window for complex state management. For efficient agentic systems, it is recommended to offload memory-intensive tasks to external

mechanisms such as Tool Calling, Model Context Protocol (MCP), or specific Agent Skills to manage state efficiently and accurately.

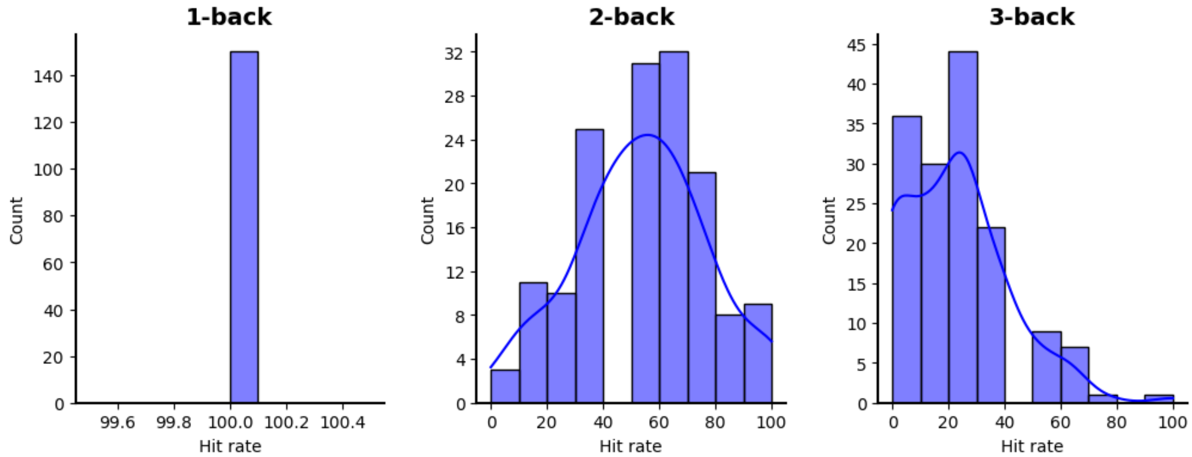
8. GitHub Repository

All codes, documentations, and experiment results are available at https://github.com/TommyS725/ChatGPT-WM#

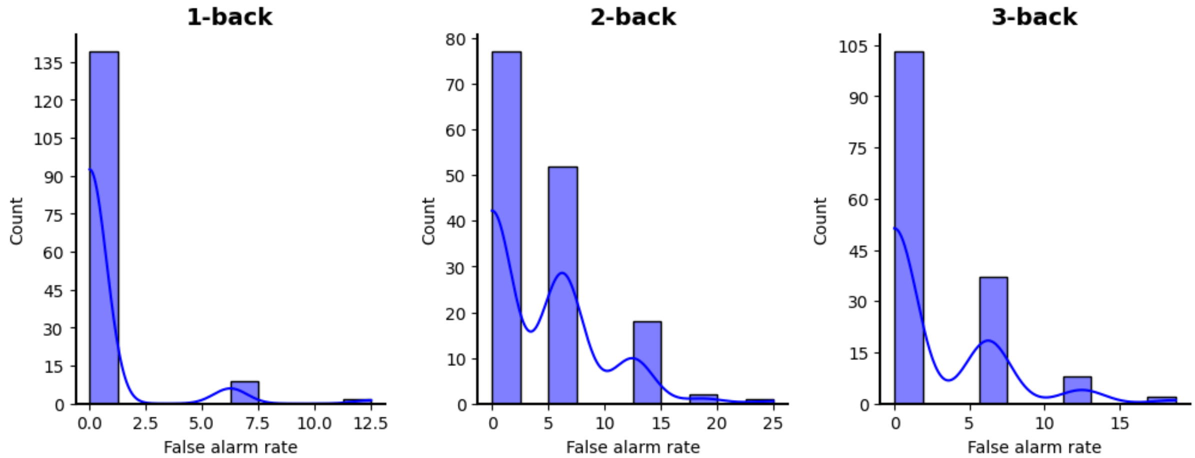
9. Appendix

Distribution Plots of Reproduced Results (deepseek-chat-v3, base verbal n-back tests)

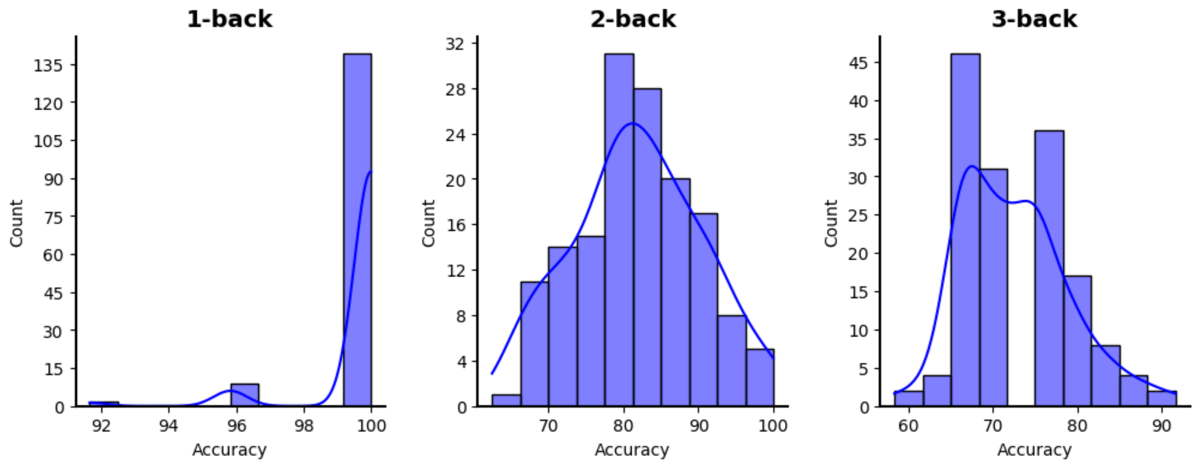
Hit Rate:



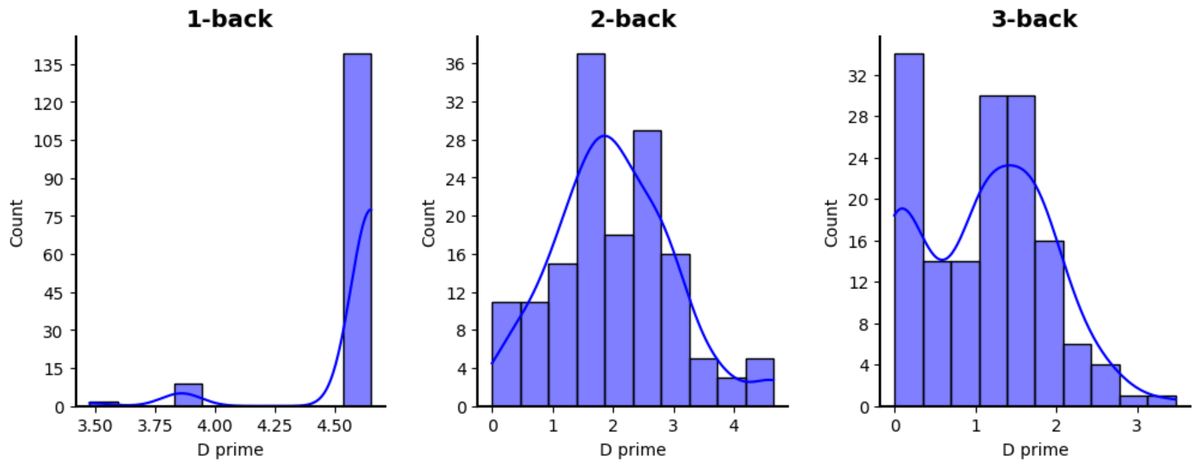
False Alarm Rate:



Accuracy:



Detection Sensitivity (d'):



Kruskal-Wallis Test Results (deepseek-chat-v3, base verbal n-back tests)

Kruskal-Wallis test for hit_rate:
H(2) = 348.8876, p = 0.0000, epsilon squared = 0.7760
Post-hoc Mann-Whitney U test:
1-back vs 2-back: U = 21825.0000, p = 0.0000, rank-biserial correlation(r) = -0.9400
1-back vs 3-back: U = 22425.0000, p = 0.0000, rank-biserial correlation(r) = -0.9933
2-back vs 3-back: U = 19031.5000, p = 0.0000, rank-biserial correlation(r) = -0.6917

Kruskal-Wallis test for false_alarm_rate:
H(2) = 62.9166, p = 0.0000, epsilon squared = 0.1363
Post-hoc Mann-Whitney U test:
1-back vs 2-back: U = 6554.5000, p = 0.0000, rank-biserial correlation(r) = 0.4174
1-back vs 3-back: U = 8540.0000, p = 0.0000, rank-biserial correlation(r) = 0.2409
2-back vs 3-back: U = 13323.5000, p = 0.0046, rank-biserial correlation(r) = -0.1843

Kruskal-Wallis test for accuracy:
H(2) = 339.3076, p = 0.0000, epsilon squared = 0.7546
Post-hoc Mann-Whitney U test:
1-back vs 2-back: U = 22028.5000, p = 0.0000, rank-biserial correlation(r) = -0.9581
1-back vs 3-back: U = 22498.0000, p = 0.0000, rank-biserial correlation(r) = -0.9998
2-back vs 3-back: U = 18058.5000, p = 0.0000, rank-biserial correlation(r) = -0.6052

Kruskal-Wallis test for d_prime:
H(2) = 326.6689, p = 0.0000, epsilon squared = 0.7263
Post-hoc Mann-Whitney U test:
1-back vs 2-back: U = 22073.0000, p = 0.0000, rank-biserial correlation(r) = -0.9620
1-back vs 3-back: U = 22499.0000, p = 0.0000, rank-biserial correlation(r) = -0.9999
2-back vs 3-back: U = 16928.0000, p = 0.0000, rank-biserial correlation(r) = -0.5047